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This document is based on the arc42 model available at https://arc42.org/overview.

1. Introduction and Goals

Requirements overview

BrokerX is a web-based application for simulated stock broking. It is an educational project that aims to replicate in a simulated environment online broking applications such as WealthSimple. No real money or personal information will be collected, exchanged, and/or used in the making, deployment and/or use of this application.

Phase 3 overview

The central goal of phase 3 was to finish implementing all the use cases and add event-driven communication. Several changes were introduced, including, but not limited to:

- New services were created, in particular for portfolios and notifications.
- Event-driven communication Kafka was implemented in the order, wallet and portfolio services for order matching (see UC07).

Quality goals

Priority	Quality goal	Scenario
1	**Maintainability**	Separation of concerns through the use of the hexagonal architecture within each service
2	**Persistence**	Support of a backend database per service with MySQL

Priority	Quality goal	Scenario
3	**Availability**	Greater or equal to 99.9% uptime
4	**Testability**	Coverage greater or equal than 80%
5	**Traceability**	Logging of errors in dedicated files
6	**Deployability**	Services must be deployable independently
7	**Scalability**	Must allow for a high number of concurent users
8	**Reliability**	Must remain operable if one or several services fail
9	**Performance**	See section Performance Reports

Stakeholders

- Developer : Learning how to design and implement a system from beginning to end.
- Clients : Exchange financial assets through a web-based interface.
- Professor/Lab assistants : Assess learning progress and competency.

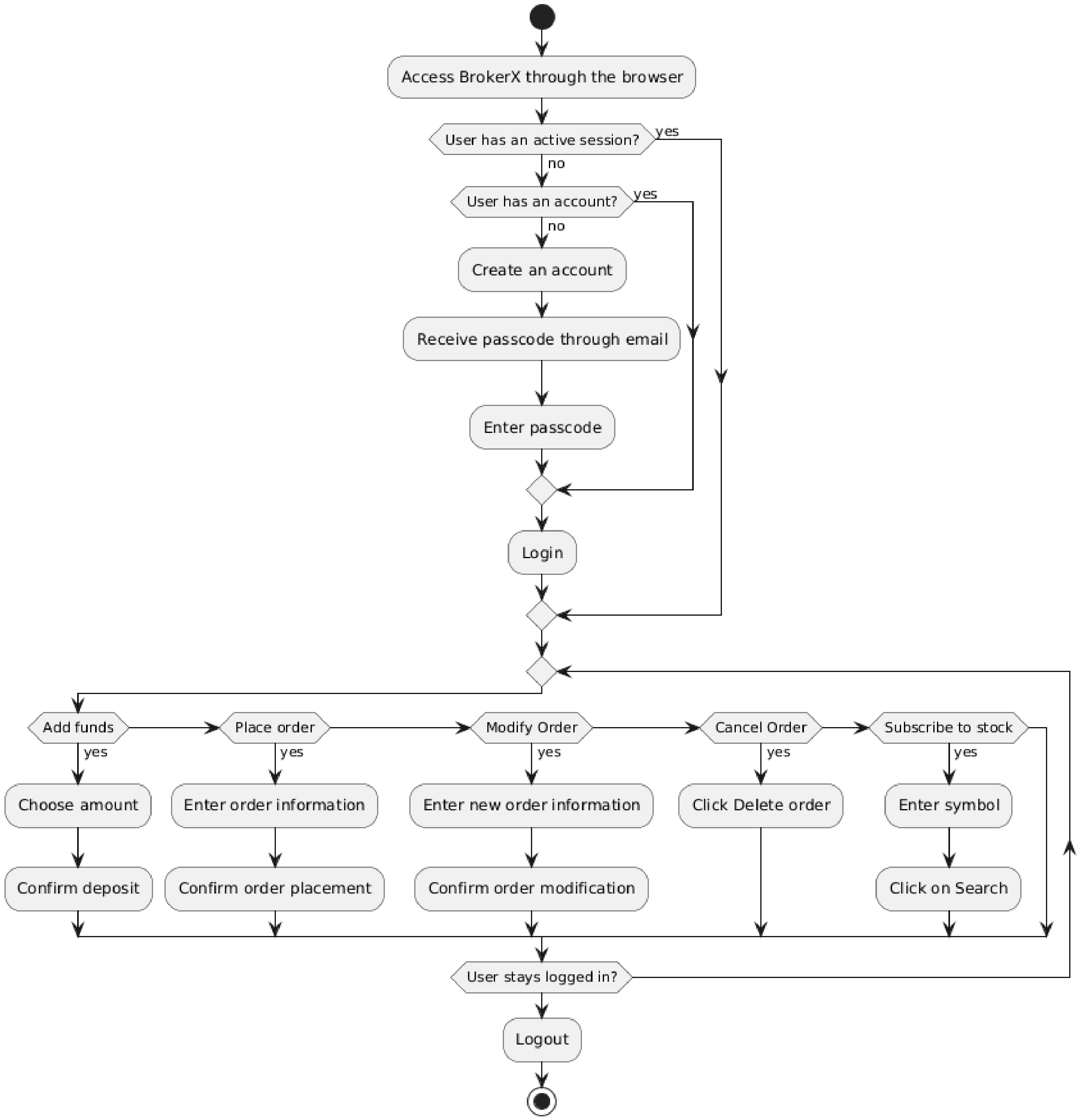
2. Architectural Constraints

Constraint	Description	Justification
Backend Technologies	Use of Python, Django, MySQL, Docker	Well documented and versatile tools
REST API	KrakenD, JWT, Postman, Swagger	Follows RESTful practices, API gateway to handle inter-service communication with authentication
Testing	Pytest (unit and integration testing), K6 (stress tests)	Reliable testing frameworks that allow full coverage and simulation of real usage
Monitoring	Prometheus, Grafana	Open-source tools with Django integration for easy monitoring and dashboards
Logging	Django built-in logging system	Integrated logging system for traceability
Caching	Redis	Fast server side caching on certain endpoints
CI/CD	Continuous integration and deployment through GitHub Actions	Ease of use, tests and deployment automation
Deployment	Deployment in Docker containers	Chosen for simplicity and portability
Event-driven communication	Asynchronous communication across services with Kafka	

3. System Scope and Context

3.1 Business Context

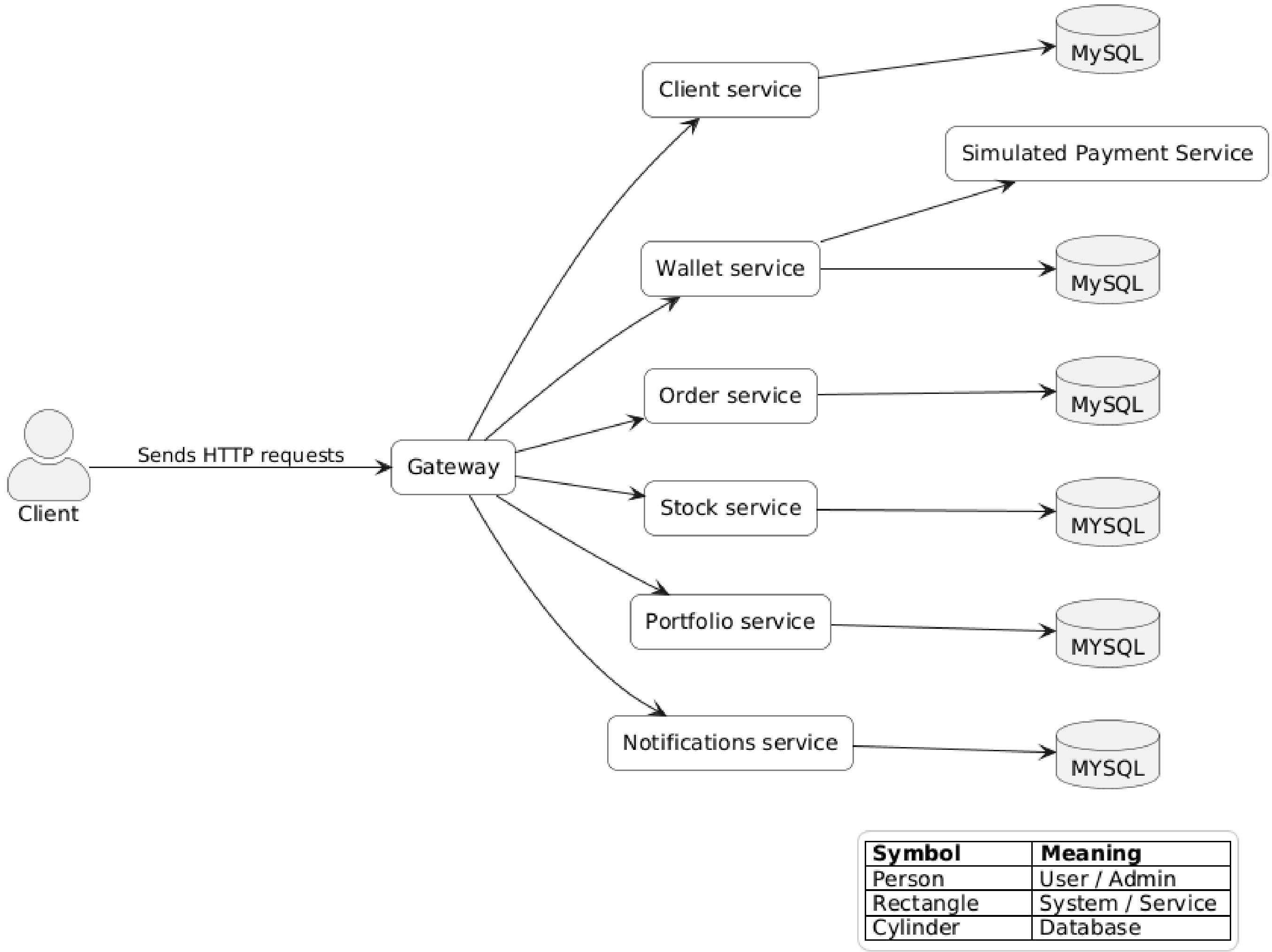
Activity diagram



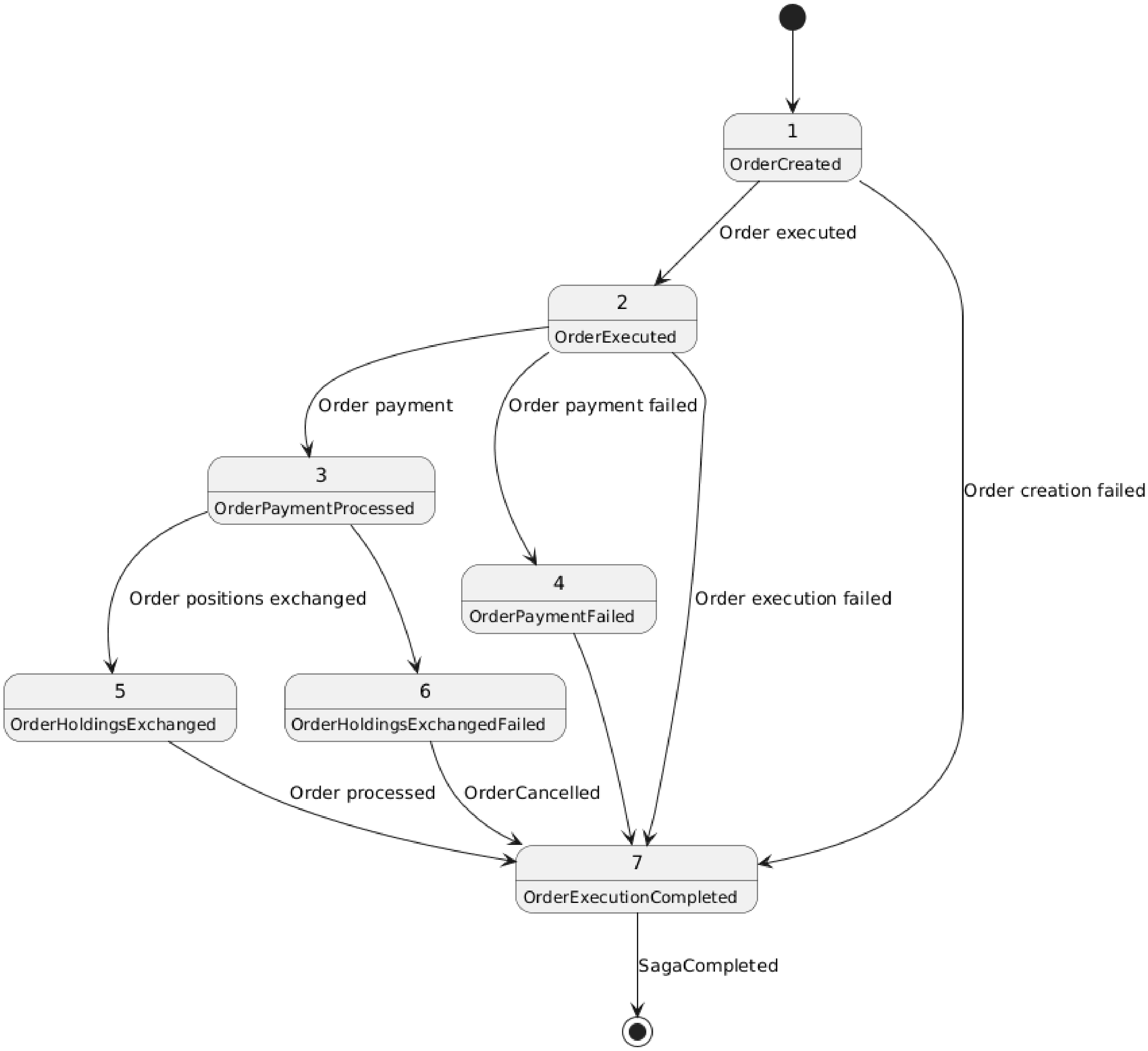
3.2 Technical Context

- **Interface** - Next.js web app with React
- **Client service** - Client registration and authentication
- **Wallet service** - Wallet funding
- **Order service** - Order placement
- **Stock service** - Stock information
- **Portfolio service** - Managing clients' assets
- **Notifications service** - Sending notifications for order executions
- **API Gateway** - Managing traffic among services
- **Persistence Layer**: MySQL databases with DAO pattern.

- **Simulated Payment Service**: Python module that mocks a banking account for clients to simulate withdrawing money from an external bank.



3.3 State machine diagram



This state machine diagram describes the choregraphed saga of UC07. Note that the diagram represents the planned implementation, in reality some states are missing from the code due to time constraints.

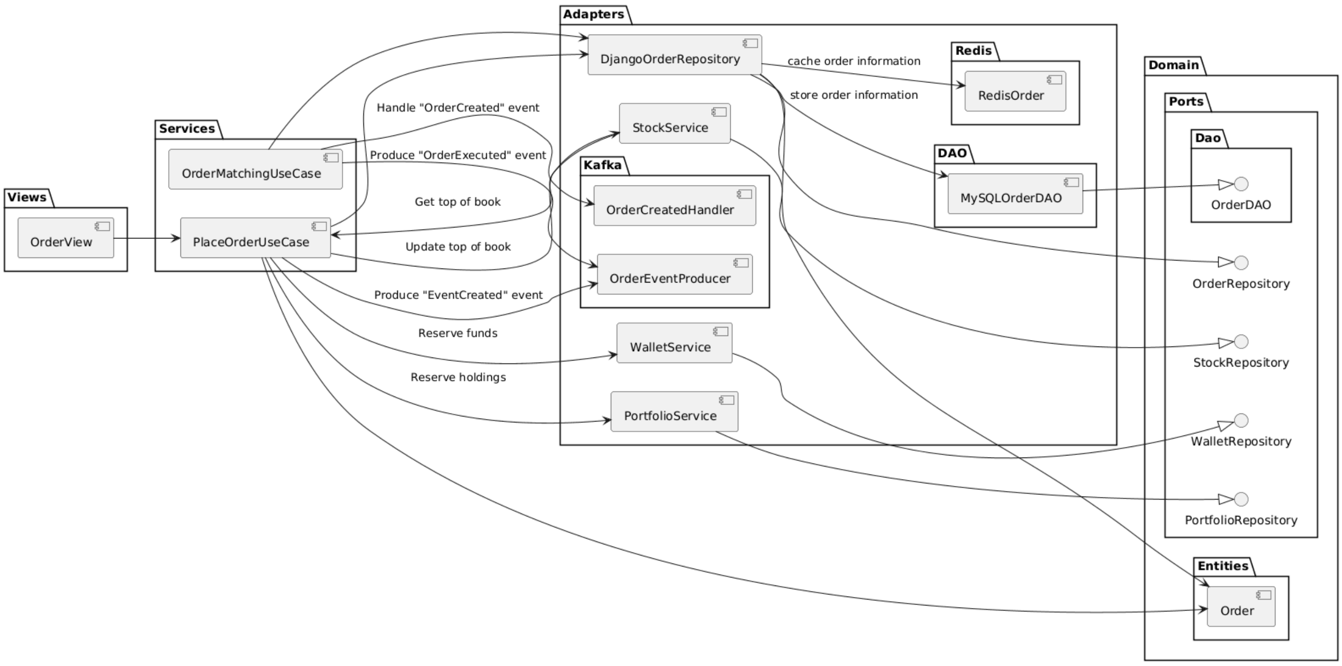
4. Solution Strategy

Problem	Solution
Separation of concerns	Microservices to isolate domain related operations and allow for partial deployability
Persistence of data	Use of the DAO pattern with Django ORM and the Data Transfer Object (DTO) pattern to communicate between the persistence layer and the service layer
Testability	Use of interfaces to simplify mocking of external sources and generation of coverage reports integrated into the CI pipeline
Maintainability	Layered architecture within each service with DTOs to pass data between layers
Integrity	Idempotency key on sensitive operations

Problem	Solution
Communication between service	API Gateway to route traffic to the appropriate service and Kafka for saga choregraphy

5. Building Block View

Component Diagram



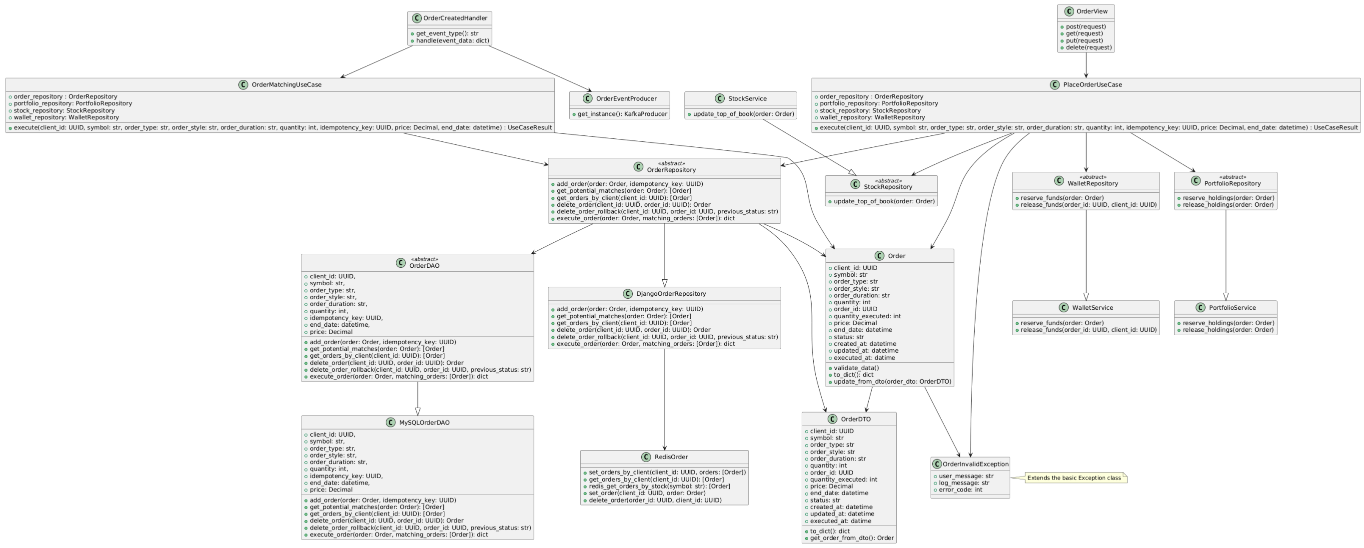
Note that this represents only one service (order) for simplicity, but unless otherwise mentioned every service follows this structure inherited from the monolithic architecture. In this architecture, the view represents an inbound adapter that directly receives requests from the user interface.

Services implement uses cases and serve to coordinate the different systems, as well as to update the view once an operation is completed.

Ports represent a contract with an external entity, whether it be part of the app, such as the DAO classes, or an external API, for example. It defines what methods and returns are expected from external entities.

Adapters implement ports and communicate with external entities through DTOs. Adapters of DAO ports are responsible for communicating with the database, while generic adapters such as `DjangoOrderRepository` coordinates between multiple entities. In general, `DjangoOrderRepository` can be thought of as an intermediary between the service layer and sources of orders, in this case the MySQL database.

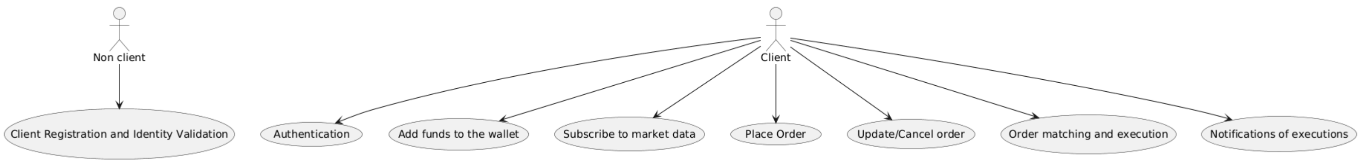
Class Diagram



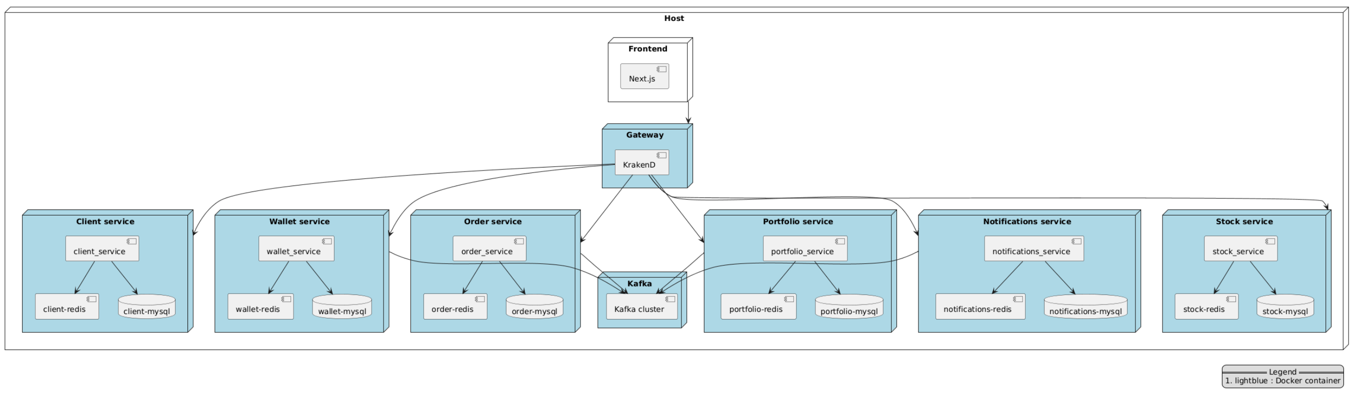
Other services follow the same pattern. Note that some utilitary classes have been avoided in orded to keep the diagram a reasonable size.

6. Runtime View

The following diagram shows the uses cases and the actors who can trigger them.



7. Deployment View



Note that in the context of the course, every service are deployed on the same host, however it is possible to deploy them on separate hosts which would allow to use more ressources and thus improve performance.

8. Cross-cutting Concepts

- DTO (Data Transfer Object) and DAO (Data Access Object) patterns
- Error and exception handling
- Logging and monitoring
- Clients

9. Design Decisions

ADR 001 – Separation of concerns with the hexagonal architecture (Phase 1)

Status

Accepted

Context

The application must be easy to maintain and extend. In particular, it must be able to evolve from a monolithic architecture into microservice-based and event-based. Moreover, testing must be facilitated and it must be easily connected to outside interfaces such as a database, external APIs, etc., without having to modify the core business logic. The different layers must be clearly separated and coupling must be reduced as much as possible. Both the hexagonal and MVC (Model-View-Controller) were considered for this purpose.

Decision

The hexagonal architecture has been chosen as the basis for the architecture of the application.

It consists of the following elements:

- Ports : Interfaces to connect the application with external sources, such as the interface layer, a database or an API.
- Adapters : Concrete implementation of a port that allows communication between the business logic and external sources without coupling the two together.

Consequences

The architectural choice will facilitate maintainability of the application as well as the transition to a microservice-based and event-driven platform. Moreover, it will reduce coupling between the different layers and simplify the addition of new functionalities and tests without having to modify the existing code. The MVC architecture has been rejected because, even though it would simplify the initial phase implementation of the platform, its tighter coupling between the user interface and the application layer would reduce maintainability and complexify future developments.

ADR 002 – Persistence of data with the DAO pattern (Phase 1)

Status

Accepted

Context

The application must reliably and securely store information regarding its clients, including the details of every transaction for personal and legal purposes. Coupling between the business layer and the persistence layer must be minimal and the system must be open to having multiple data sources, for instance databases and CSV files.

Decision

To access the MySQL database, the DAO (Data Access Object) will be used as a way to lessen coupling between the business layer and the persistence layer. Moreover, an ORM will be used within the DAO in order to streamline and simplify database queries. Both the DAO pattern and the ORM were in contention for the above purposes, however ultimately it has been decided that since they offer complementary roles, a combination of both yields the best results in the context of this application.

Consequences

This choice for the persistence layer will isolate the data from the business layer and provide an abstraction to communicate with the database, namely a data access object interface. This decision will minimize coupling between both layers and provide a simple interface to make queries to the database. As such, it will promote high cohesion, facilitate unit testing by allowing for database mocking, and protect the business layer from modification and/or extension of the database model.

ADR 003 – Implementation in Python with the Django Framework (Phase 1)

Status

Accepted

Context

The application needs a platform to be developed on. After initial brainstorming, two options were left, Django or Java with SpringBoot. Considerations were made for performance, level of previous knowledge, ease of use, and more. This is a major decision that will impact the application for the rest of its life cycle, and is something that impacts how the application is conceived, tested, maintained and deployed.

Decision

The Django framework has been chosen as the basis of this project.

Django is a popular and very well-documented framework, which played a part in this decision. Moreover, it offers various services such as an authentication system, an ORM, a logging system, and more. The convenience of having such features bundled in a single framework made this an attractive option. Furthermore, Django allows easy and traceable database migrations, which factored in the benefits of using it. Additionally, the fact that I am familiar with Django and have no experience with SpringBoot had a role in this decision.

Consequences

The choice of framework will impact how the application is designed. The main drawback of Django is its inferior performances when compared to Java, however I believe that with smart caching and other techniques, the quality requirements for the project will be able to be met and even exceeded with Django.

ADR 004 – Caching with Redis

Status

Accepted

Context

Caching is an essential strategy to improve latency on common endpoints. It allows to bypass database queries on GET endpoints which are costly operations, as well as quickly failing if two many connections are set at once. Therefore, caching lightens the load on the database which makes the whole application faster and more reliable

Decision

Redis as been chosen as the backend cache.

Redis is a popular and fast key-value database used as a caching mecanism for backends. It has been chosen for this project for its usability with simple and easily readable syntax, its readily available documentation and fast performance. A redis instance will be deployed for every service that needs caching and will use a write-through invalidation method, in other words, when a value is written to the database, it will also be written to Redis, ensuring a high level of cohesion between the two.

Consequences

Redis will improve the performance of the application and be used everywhere a query is made to the database. To ensure data integrity between Redis and the database, it is essential to update both at the same time.

ADR 005 – KrakenD API Gateway for communication between services

Status

Accepted

Context

Services must be able to communicate with each other, however we want to avoid front end users interacting directly with the services, there we must put an API layer in front. This will allow to have control over the requests that are sent to the services.

Decision

KrakenD as been chosen as the API gateway.

KrakenD will provide the API Gateway to arrange traffic between services. It has been chosen its simple configuration and customization which allows for authorization with JWT tokens, caching and load balancing.

Consequences

Requests will now all have to be routed to the API gateway rather than directly to the backend. This means that the gateway must correctly direct HTTP requests to the appropriate service and transmit responses.

10. Quality Requirements

Performance

- Throughout of ≥ 800 ordres/s on orders
- P95 latency ≤ 250 ms

Testability

- Coverage for unit tests greater or equal to 80%
- Full unit, integration and E2E testing
- Automated coverage report in CI

Reliability

- No failure in case of a system error
- Logging of errors

Deployability

- Possibility of deploying only some of the services

11. Risks and Technical Debt

- The migration to microservices comes with a higher level of complexity
- Using Django might lead to performance issues.
- MySQL with the Django ORM might struggle under a high load.

Mitigation Strategies

- High coverage testing to detect bug as they are created.
- Caching and load balancing to improve performance

12. Glossary

Term	Definition
Hexagonal Architecture	An architecture model that isolates the core business logic from external entities
Client	An individual that uses the platform and any number of its functionalities.
User	Used strictly for Django authentication. Not to confuse with Client, which is a part of the DDD. The user is simply the default Django model to authenticate clients
(Digital) Wallet	Capital that a client has available in their account to make transactions.
Withdrawal	Adding funds to a client's wallet.
Auditing	Examination of financial assets and operations of an entity for legal and management purposes.
Simulated Payment System	Represents an external source from which clients can withdraw money (e.g., a bank).
Transfer	Operation in which financial assets or funds are exchanged.

Term	Definition
DAO (Data Access Object)	Pattern that aims to isolate database queries
DTO (Data Transfer Object)	Pattern that allows passing data between layers
Microservices	An architectural paradigm in which domain related operations are separated into distinct, independent units
API Gateway	Allows for communication with microservices
Monitoring	Data collection for performance analysis of a software program

Performance reports

The following table presents key metrics that were polled at different stages of the application's lifecycle. Metrics were taken with 150 concurrent user.

Methodology

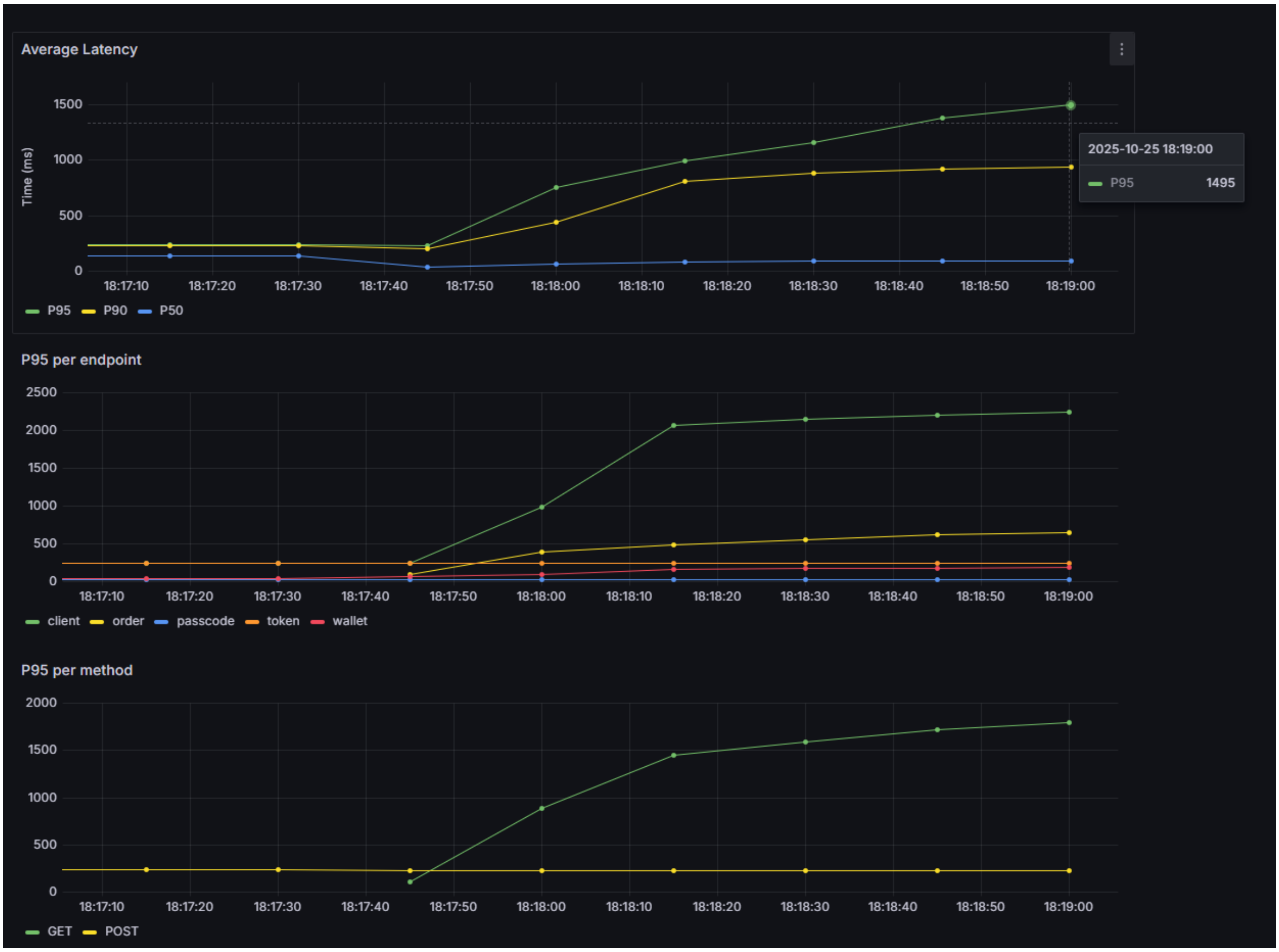
The stress tests use `k6` to simulate HTTP traffic to `broker-app:8000`. The set up phase creates 150 users and keeps their tokens in an array. Then, virtual users randomly select a token and call the specified endpoints with this token. This ensures that as virtual users loop through the calls, they do not create more than 150 clients. Client registration and passcode validation are rare events, thus including them at a similar rate to common operations skews results. The tests ramp up to 150 users in 15 seconds, then remained steady for a 1 minute, then have a 30s rampdown.

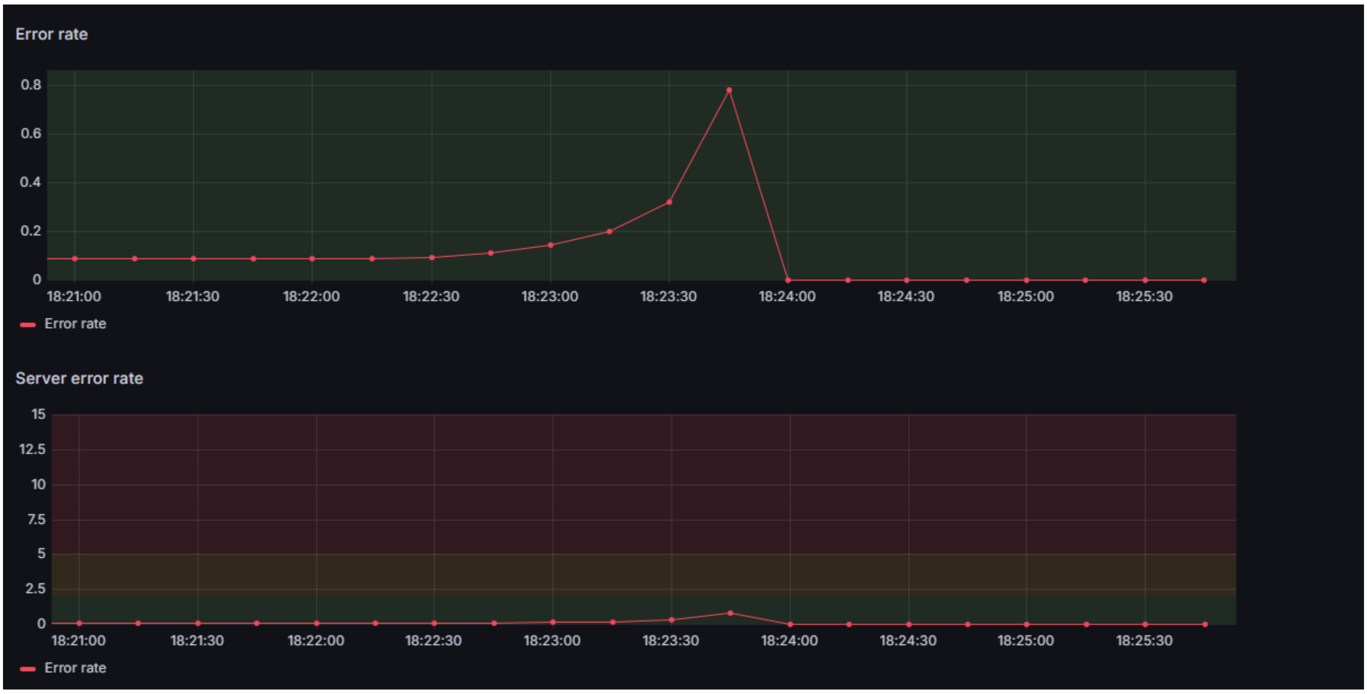
Because of time constraints, performance for the microservices with event-driven communication was not yet reported, but should hopefully be done by the deadline.

Metric	Monolith	Monolith + Redis
P95	1495ms	468ms
P90	936ms	429ms
P50	95.5ms	175ms
Throughput (orders/s)	81.3	120
P95 on POST /order	649ms	475ms
Max concurrent users (error rate < 5%)	150	170

Observations

Monolithic





Monolithic + Redis

The addition of Redis significantly improved performance on the app, decreasing the P95 by 319%. We can see through the Redis CLI that more than 90% of requests to Redis hits the cache, in other words 90% of requests that previously went to the database are now being handled by Redis, which explains the spike in performance. Redis however only serves fetch operations, the others thus need to be a focus for further optimization.

Runbook

Overview

This runbook provides operational procedures and troubleshooting steps for managing BrokerX.

Running with `deploy.sh` script

1. To start the service, begin by cloning the Github repository available at https://github.com/William-Lavoie/log430-a25-labo5 or use the ZIP file with the source code.
2. Run `BrokerX/deploy.sh` with the name of the services you want to run (i.e `BrokerX/deploy.sh client`) to launch only the client service. If you do not specify parameters, every service will be started. Note that this is what you need to do to follow the demo guide.
3. The frontend application should now be running on port 3000 (`http://localhost:3000/`)
4. You can access the Swagger documentation at `http://localhost:8001/api/`

Running manually

If you prefer not to use the script or it does not work, you can follow these steps.

1. Go to the directory of the service you wish to run (e.g `client_service`)
2. Run `docker compose down -v` (or `docker compose down` if you want previous data to persist)
3. Run `docker compose build`
4. Run `docker compose up` (or `docker compose up -d` if you want it running in the background)
5. Follow steps 3-5 in the `gateway` directory.
6. Go to `react_frontend`
7. Run `npm install`
8. Run `npm run dev`

Diagnosing errors

Errors are automatically logged in the various log files under the `log` directory in each service. Every service has a separate log files for http requests, mysql, redis and general messages.

Accessing the database for a service

You can access MySQL command line as root by running the following command in the service's directory: `docker exec -it {service}-mysql mysql -u root -p`

Running tests

You can run the tests by running the command in the service's directory: `docker exec {service-app} python -m pytest`

Demo Guide: Phase 3

1. Create an account

1. Navigate to the root page `/`. Since you are not logged in, you will be redirected to `/login`.
2. Click on the link at the bottom to go to the account creation page (`/create_user`).
3. Enter your information.
4. Click "Submit". If any of the information your entered is not correct, you will be notified.
5. You will be redirected to `/login`.

2. Login and verify your passcode

1. Enter your login information (email and password).
2. Click on Sign in.
3. You will be redirected to `/create_account/validate_passcode`. Note that there might be a delay since you are first redirected to `/`.
4. A passcode has been sent to you by email, however since no SMTP server is configured at the moment, the passcode is sent through logs. Please look at `BrokerX\client_service\logs/client_logs` to know your passcode. You can also request a new one if it doesn't work.
5. Enter your passcode.
6. You will be redirected to `/`.

3. Add funds to your wallet

1. In the navbar, click on "Wallet".
2. Your current balance will be shown. It should be 0$.
3. Enter the amount you wish to add.
4. Click on "submit"
5. The amount is automatically increased. Note that there is a 10,000$ limit on wallets. Please note that since the payment system is simulated, you can always withdraw at most 1000$ at a time, however specific value (10.0, 20.0, 30.0, 40.0, 50.0, 60.0, 70.0, 80.0) trigger errors as a way to tests potential errors coming from the external payment service.

4. Subscribe to market data

1. Enter the symbol for which you wish to subscribe to market data in the searchbar in the navbar. By default AAPL and XEQT are present in the database so you can try those.
2. Click on search
3. You should be redirected to the

5. Place Order

1. Click on Place Order in the navbar.
2. You should be redirected to `/place_order`.
3. Enter the information
4. Click on submit
5. The order should appear below.

Note that if you select Limit and/or GTD in the duration input, additional input fields will appear.

6. Modification/Cancellation of orders

1. Navigate to `/place_order` if you are not already there from step 5.
2. Create an order if you do not have one.
3. Click on Edit.
4. Enter the new information.
5. Click on Submit.
6. You should now see the new order.
7. Press on Delete
8. You should no longer see the order

You can create an order using the symbol XEQT to make sure that it won't be executed before you can modify it.

7. Execution of order

This UC does not have a test guide, it should automatically occur when you do step 5. Use AAPL as it is the only symbol with pre set orders in the database.

Confirm that the information has been updatedin the wallet page, order page and portfolio page.

8. Notification of execution

This UC is not yet functional, but I hope to get it done by the final deadline.

Additional Notes

- Your session will remain active until you log out manually or the session expires.
- If any redirect fails, please manually navigate to the specified page.

Phase 3 Retrospective

There are several things that could not be done in time for the end of the project. Next steps would be finishing the sage for UC07, fixing the implementation of UC08 and adding add balancing, which was present before migrating to microservices.

If I were to do this project again, I would do many things differently, for example I would spend less time working on the UI, I would have migrated to microservices earlier and I would split the programs in fewer services since it has proven to be very difficult to coordinate between them, and as a result test coverage has dropped, the CI/CD pipeline is no longer operational and neither is monitoring, even tho all those items were properly set up at one point. Overall, I am satisfied with the result but given additional time there are many improvements I would do. As is, the program is somewhat shaky and there may be several undocumented bugs and issues.

MoSCoW Prioritization Table – Use Cases

Use Case ID	Use Case Title	Priority	Justification

Use Case ID	Use Case Title	Priority	Justification
UC-01	User Registration	Done	Necessary for clients to access the platform.
UC-02	Authentication and MFA	Done	Clients need a secure and reliable way to log in while protecting their identity and personal information.
UC-03	Adding Funds to Wallet (Virtual Deposit)	Done	Clients need funds to place buy orders.
UC-04	Subscription to Market Data	Must	Allows clients to consult real-time quotations to make informed decisions without reliance on an external information source.
UC-05	Placing Buy/Sell Orders	Done	Core feature that clients sign up for.
UC-06	Modification/Cancellation of Orders	Must	Gives clients flexibility and peace of mind.
UC-07	Internal Matching and Execution	Done	Transactions between buyers and sellers cannot be completed without this.
UC-08	Confirmation of Executions and Notifications	Should	Nice to have for transparency and client trust.

References

- This project has been made in collaboration with chatGPT for the purposes listed below. Note that uses of artificial intelligence in this project is limited to strictly those listed below.
    - Creation of templates for use cases, glossary, MoSCoW prioritization table and arc42.
    - Generation of some CSS/HTML/Javascript code for the frontend
    - Asking if the artifacts satisfy the requirements.
    - Used as a search engine.
    - Improve readability (phrasing, grammar, spelling, etc)
    - Debugging

- Petrillo, F. (2025, Fall). Class notes [pdf]. LOG430, École de Technologie Supérieure.

- Ullman, G. (2025, Fall). Github labs. LOG430, École de Technologie Supérieure.

- Documentation for all the dependencies listed in requirements.txt as well as any technology mentionned in this document